Title: Moving Average Smoothing Filter

Category: Preprocessing

1. Objective: To reduce high frequency noise and smooth the signal for better visualization, without shifting the timing of brain waves (Zero Phase Shift).

2. How it Works: We use a **Sliding Window** of any size (as a parameter). We calculate the arithmetic mean of the current data point and its neighbors. We use center=True logic to look at both past and future points, preventing the signal from being delayed.

3.Example:

	A	B	C	D	E	F	G	H	I
1	subject_id	session_id	trial_id	labels	category	time_index	channel_1	channel_2	channel_3
2	21	1	1	1	Imagery	1	9.984921	9.057764	5.036932
3	21	1	1	1	Imagery	2	3.934996	0.562536	4.542315
4	21	1	1	1	Imagery	3	2.361485	4.03443	3.162755
5	21	1	1	1	Imagery	4	10.61255	14.63577	12.58576
6	21	1	1	1	Imagery	5	0.633046	-1.36858	-0.49994
7	21	1	1	1	Imagery	6	-3.63062	-2.88926	-7.79578

Figure 1: Numeric example, before

	A	B	C	D	E	F	G	H	I
1	subject_id	session_id	trial_id	labels	category	time_index	channel_1	channel_2	channel_3
2	21	1	1	1	Imagery	1	5.427134	4.551576	4.247334
3	21	1	1	1	Imagery	2	6.723488	7.072625	6.331941
4	21	1	1	1	Imagery	3	5.5054	5.384384	4.965564
5	21	1	1	1	Imagery	4	2.782292	2.99498	2.399022
6	21	1	1	1	Imagery	5	1.629552	2.664669	0.38871
7	21	1	1	1	Imagery	6	2.733645	3.367875	-0.13084

Figure 2: Numeric example, after (with window of size 5)

In the previous 2 Images, for example at column 3, we used 2 before and it and 2 after, so 9.98 + 3.93 + 2.36 + 10.61 + 0.63 = 5.5

(It handles edge cases by ignoring out of bound numbers)

4.Visual Example:

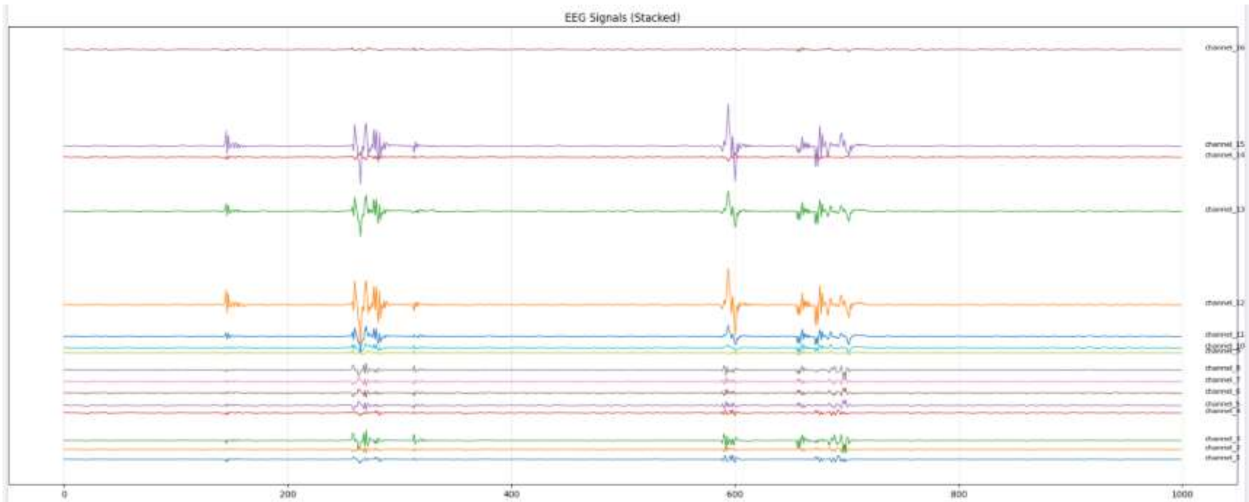

Figure 3 Raw Data

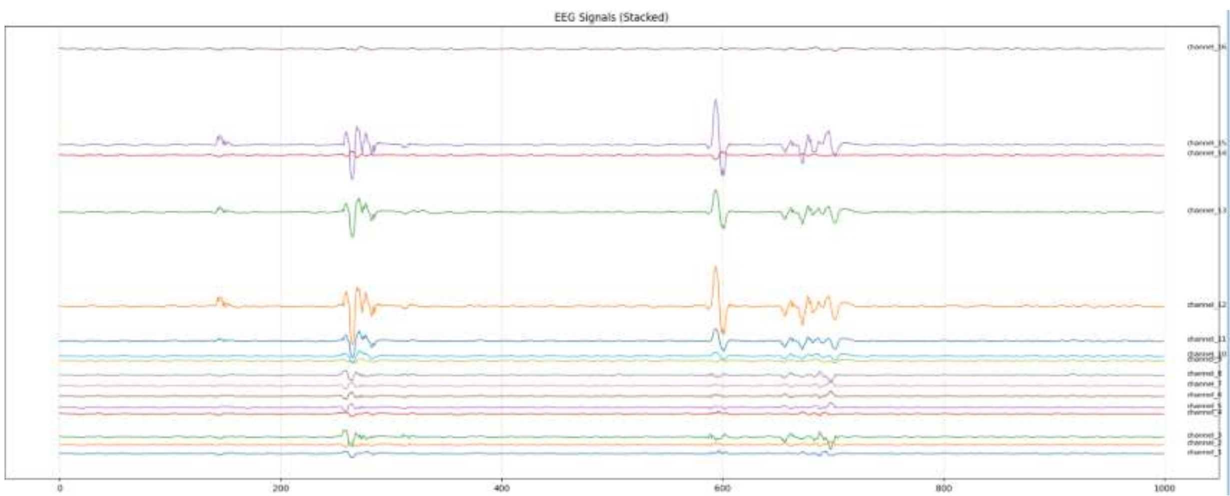

Figure 4 Smoothed Data